

Critical Scaling in Hyperbolic Attention Mechanisms: *Extended to Incorporate Curvature, Topological Dependence, and Fractal Dimension*

William Chuang

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Abstract

We provide an extensive, originally 20-page exposition of a hyperbolic self-attention framework for transformers, placing it on a rigorous mathematical foundation. This paper devotes its content to formal derivations, equations, and symbolic representations, reflecting the complexity of critical phenomena, fractal hyperbolic geometry, and advanced attention mechanisms. We present, in detail, how the critical inverse temperature $\beta_c(\delta)$ depends on the fractal dimension δ , derive spectral densities in hyperbolic spaces, and outline an actionable implementation for dynamic scaling in transformers. **Furthermore, we extend our discussion to more general curved manifolds (with variable curvature κ), exhibit how topological connectivity $\mathcal{T}$ among neurons can shift the critical threshold, and unify these factors with fractal dimension δ .** Our approach integrates both theoretical constructs (e.g., Laplace–Beltrami operators, monodromy arguments, and partition functions) and practical algorithmic protocols (e.g., minimization of energy dissipation, complexity analyses). Readers well-versed in modern geometry and statistical mechanics will find a thorough, self-contained development of hyperbolic (or generally curved) attention and its consequences for large-scale machine learning.

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1 Introduction

The success of transformer-based architectures in natural language processing (NLP), computer vision, and other domains has been largely driven by the self-attention mechanism introduced by Vaswani et al. in the seminal work *Attention is All You Need* [3]. While standard dot-product attention with a scaling factor $\frac{1}{\sqrt{d_k}}$ has proven effective, theoretical connections to statistical mechanics and hyperbolic geometry have remained underexplored. Recent research [1] suggests that we can interpret the softmax normalization as a Boltzmann distribution with an inverse temperature β , highlighting analogies to Ising spin systems and critical phenomena.

In parallel, hyperbolic spaces have emerged as powerful representation tools for hierarchical data, large-scale graphs, and fractal-like structures. The combination of hyperbolic embeddings with self-attention, therefore, poses an intriguing avenue for harnessing geometric curvature, fractal dimensions, and critical scaling in next-generation transformers.

Paper Overview. The goal of this paper is to develop a *mathematically rigorous* and fully *self-contained* treatment of hyperbolic attention mechanisms near critical points. We expand upon preliminary outlines to provide explicit derivations, bridging fractal geometry, spectral theory, and large-scale optimization. Sections 2–3 build essential background. Section 4 introduces the notion of criticality, and Section 5 rigorously derives $\beta_c(\delta)$ as a function of δ . Sections 6–7 propose an actionable protocol for integrating spectral feedback and dynamic scaling. Later sections explore numerical experiments, quantum security, and extended directions.

Extended Scope. *Beyond the original hyperbolic case, we incorporate negative curvature κ , topological connectivity $\mathcal{T}$ of neural units, and fractal scaling. We demonstrate that the critical inverse temperature β_c can in fact be approximated by a more general function*

$$\beta_c(\delta, \kappa, \mathcal{T}) \approx \left(\lambda_{\max}(\mathcal{T}) \right)^{-1} \exp\left(C(\kappa) \delta r_{\text{eff}} \right),$$

where $\lambda_{\max}(\mathcal{T})$ is the spectral radius reflecting neuron connectivity and $C(\kappa)$ captures curvature effects.

Intended Audience. This manuscript assumes familiarity with hyperbolic geometry (e.g., the Poincaré disk, upper half-plane, or Minkowski models), basic knowledge of spectral theory, and a working understanding of transformer architectures.

2 Preliminaries and Notations

In this section, we define our notational conventions and briefly review essential ideas in hyperbolic geometry and fractal scaling. We also summarize the structure of standard self-attention in transformers. **Later, in Section 8, we extend these notions to more general Riemannian manifolds with curvature $\kappa(x)$ and adjacency topologies $\mathcal{T}$.**

2.1 Hyperbolic Geometry and Fractal Structures

A hyperbolic manifold $\mathbb{H}^n$ can be represented in several equivalent models:¹ for instance, the Poincaré disk model $(\mathcal{D}, g_{\mathcal{P}})$ is defined as

$$\mathcal{D} = \{x \in \mathbb{R}^n : \|x\| < 1\}, \quad \text{with metric} \quad ds^2 = 4 \frac{\|dx\|^2}{(1 - \|x\|^2)^2}.$$

Fractal scaling, encountered in hierarchical or tree-like data, can often be captured by the Hausdorff dimension or other fractal dimensions. We adopt the fractal dimension parameter δ introduced in eq. (1).

Definition 1 (Fractal Dimension Parameter). *Let $m \in \mathbb{Z}^+$ denote a branching factor and r_{eff} be an effective radius in the hyperbolic manifold. We define[2]:*

$$\delta := \frac{\log(2m - 1)}{r_{\text{eff}}}. \tag{1}$$

In Section 5, δ plays a central role in determining the critical inverse temperature $\beta_c(\delta)$.

¹The Poincaré disk model, the Poincaré half-plane model, or the hyperboloid (Minkowski) model.

2.2 Transformer Architectures and Self-Attention

Standard self-attention, given an input sequence $X \in \mathbb{R}^{N \times d}$, employs query, key, and value projections $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$. The (scaled) dot-product attention weights are

$$A_{ij} = \frac{(XW_Q)_i (XW_K)_j^T}{\sqrt{d_k}}, \quad Z = \text{Softmax}(A) V. \quad (2)$$

Hyperbolic variants of eq. (2) replace the Euclidean dot-product with hyperbolic inner products, as discussed in Section 3.

3 Mathematical Formulation of Hyperbolic Attention

3.1 Generalized Hyperbolic Attention Operator

Consider a map $\phi : \mathbb{R}^d \rightarrow \mathbb{H}^n$ embedding Euclidean input vectors $(XW_Q)_i$ and $(XW_K)_j$ into an n -dimensional hyperbolic manifold $\mathbb{H}^n$. Then define the hyperbolic attention operator:

$$A_{ij}^{(\mathbb{H})} = f_\kappa(\langle \phi((XW_Q)_i), \phi((XW_K)_j) \rangle_{\mathbb{H}}), \quad (3)$$

where $f_\kappa(\cdot)$ is a function ensuring suitable scaling/normalization, and $\langle \cdot, \cdot \rangle_{\mathbb{H}}$ is the hyperbolic inner product. In practice, one might take $f_\kappa(x) = \kappa x$ or a variant that ensures stable magnitudes.

3.2 Spectral Representation

We consider the Laplace–Beltrami operator $\Delta_{\mathbb{H}}$ on $\mathbb{H}^n$. Its eigenfunctions $\{\psi_n\}$ and eigenvalues $\{\xi_n\}$ satisfy

$$\Delta_{\mathbb{H}} \psi_n = -\xi_n \psi_n, \quad \xi_n \geq 0. \quad (4)$$

The spectral density $\rho(\xi)$ encodes the distribution of eigenvalues in the hyperbolic manifold. Under fractal/hierarchical geometry, we often encounter power-law behavior

$$\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1} \quad (\xi \rightarrow 0), \quad (5)$$

linking the fractal dimension parameter δ to long-range correlations in the manifold.

4 Critical Phenomena in Hyperbolic Transformers

4.1 Definition of Criticality in Attention Mechanisms

We now interpret the hyperbolic inner product $\langle \phi((XW_Q)_i), \phi((XW_K)_j) \rangle_{\mathbb{H}}$ as an Ising-like spin alignment. Concretely, define:

$$S_i := \phi((XW_Q)_i), \quad S_j := \phi((XW_K)_j), \quad \text{such that} \quad \langle S_i, S_j \rangle_{\mathbb{H}} \equiv S_i S_j.$$

Hence, the energy contribution from each pair (i, j) can be expressed similarly to an Ising Hamiltonian:

$$H = - \sum_{\langle i, j \rangle} \langle S_i, S_j \rangle_{\mathbb{H}} = - \sum_{\langle i, j \rangle} S_i S_j. \quad (6)$$

In statistical physics, one introduces the partition function:

$$\mathcal{Z}(\beta) = \sum_{\{S\}} e^{-\beta H}. \quad (7)$$

A system is said to be critical if

$$\frac{\partial^2}{\partial \beta^2} \log \mathcal{Z}(\beta) \Big|_{\beta=\beta_c(\delta)} \rightarrow \infty, \quad (8)$$

indicative of long-range correlations and divergent susceptibilities.

4.2 Fractal Scaling Parameter

We reiterate eq. (1):

$$\delta = \frac{\log(2m-1)}{r_{\text{eff}}}, \quad (9)$$

where m is the branching factor and r_{eff} is the effective radius of the hyperbolic manifold. The exponent δ controls the curvature-induced fractal dimension, shaping the entire spectral structure of $\Delta_{\mathbb{H}}$.

5 Explicit Derivation of $\beta_c(\delta)$

5.1 Partition Function and Hamiltonian

Combining eqs. (6) and (7), we note:

$$\mathcal{Z}(\beta) = \sum_{\{S\}} \exp\left(\beta \sum_{\langle i,j \rangle} S_i S_j\right), \quad (10)$$

$$H = - \sum_{\langle i,j \rangle} S_i S_j. \quad (11)$$

One typically isolates β as a free parameter, scanning from low (disordered) to high (ordered) inverse temperature.

5.2 Spectral Density and Critical Eigenvalues

From eq. (5), we approximate near $\xi = 0$:

$$\rho(\xi) \approx C |\xi|^{\frac{\delta}{2}-1}, \quad C > 0, \quad \xi \in [0, \xi_{\text{max}}]. \quad (12)$$

Here, δ enters as a measure of fractality. The correlation length diverges at criticality, implying certain zero modes $\xi \approx 0$ become dominant.

5.3 Critical Inverse Temperature Derivation

We define the second derivative of the free energy $F(\beta) = -\log \mathcal{Z}(\beta)/\beta$:

$$\frac{\partial^2}{\partial \beta^2} \log \mathcal{Z}(\beta) = \text{Var}_{\beta}(H) = \langle H^2 \rangle_{\beta} - \langle H \rangle_{\beta}^2. \quad (13)$$

Criticality requires $\text{Var}_\beta(H) \rightarrow \infty$. In hyperbolic geometry, the relevant variance can be mapped to integrals over ξ (the spectral domain). Setting this variance to diverge yields:

$$\int_0^{\xi_{\max}} \frac{\rho(\xi)}{(\beta\xi - 1)^2} d\xi \longrightarrow \infty \quad \text{as} \quad \beta \rightarrow \beta_c(\delta). \quad (14)$$

Near $\xi = 0$, eq. (12) indicates the integral diverges if $(\beta\xi - 1)$ vanishes at $\xi = \xi_{\text{crit}}$. We identify $\xi_{\text{crit}}(\delta) \sim 1/\beta_c(\delta)$, matching typical arguments in statistical mechanics. Coupled with the hyperbolic spectral gap scaling, $\xi_{\text{crit}} \sim e^{-2\delta r_{\text{eff}}}$, we arrive at:

$$\frac{1}{\beta_c(\delta)} \sim e^{-2\delta r_{\text{eff}}} \implies \beta_c(\delta) \sim e^{2\delta r_{\text{eff}}}. \quad (15)$$

Since $\delta = \frac{\log(2m-1)}{r_{\text{eff}}}$, we substitute:

$$\beta_c(\delta) \sim e^{2 \frac{\log(2m-1)}{r_{\text{eff}}} r_{\text{eff}}} = (2m-1)^2. \quad (16)$$

Thus, we obtain the explicit relationship:

$$\boxed{\beta_c(\delta) \approx (2m-1)^2, \quad \text{where} \quad \delta = \frac{\log(2m-1)}{r_{\text{eff}}}.} \quad (17)$$

6 Spectral Feedback and Dynamic Scaling

6.1 Spectral Feedback Parameter $\varepsilon(\delta)$

We incorporate a *spectral feedback parameter* to align the attention update rule with hyperbolic geometry. Let $\{\xi_n\}$ be the eigenvalues from eq. (4) and define:

$$\varepsilon(\delta) \sim |\xi|^{-\frac{\delta}{2}}, \quad (18)$$

so that large eigenvalues are re-weighted less aggressively than small ones, in line with fractal spectral weighting.

6.2 Minimization of Energy Dissipation

Define an energy dissipation functional:

$$E_{\text{diss}}(\beta, \varepsilon) = \sum_{i,j} |A_{ij}^{(\mathbb{H})}(\beta, \varepsilon) - A_{ij}^{\text{optimal}}|^2, \quad (19)$$

where A_{ij}^{optimal} might represent a target or reference distribution. We look for $\frac{\partial E_{\text{diss}}}{\partial \varepsilon(\delta)} = 0$. Using eqs. (17) and (18), we see how $\varepsilon \rightarrow \varepsilon(\delta)$ ensures minimal energy consumption while preserving the essential correlation structure.

7 Hyperbolic Attention Implementation

7.1 Algorithmic Protocol

Algorithm 1: Dynamic Hyperbolic Attention

1. **Input:** X, W_Q, W_K, W_V , hyperbolic embedding ϕ , branching factor m , radius r_{eff} .
 2. **Compute:** $\delta \leftarrow \frac{\log(2m-1)}{r_{\text{eff}}}$.
 3. **Set:** $\beta \leftarrow \beta_c(\delta) \approx (2m-1)^2$.
 4. **Compute:** $\varepsilon(\delta)$ using eq. (18).
 5. **Construct:** Hyperbolic attention matrix $A_{ij}^{(\mathbb{H})} \leftarrow f_{\kappa}(\langle \phi((XW_Q)_i), \phi((XW_K)_j) \rangle_{\mathbb{H}})$.
 6. **Apply:** $Z \leftarrow \text{Softmax}(A^{(\mathbb{H})})(XW_V)$.
 7. **Update:** Minimizing E_{diss} w.r.t. ε if necessary.
- Output:** Updated hyperbolic attention outputs Z .

7.2 Computational Complexity

Hyperbolic distance calculations typically require $O(n)$ for n -dimensional embeddings. The spectral weighting can introduce overhead scaling with the number of eigenvalues N_{ξ} if a direct spectral transform is used. However, approximate factorization or randomization can reduce this complexity to $O(N \log N)$ or $O(N)$, depending on the geometry.

8 Extended Curvature and Topological Dependence

We now *extend* the hyperbolic model to a more general Riemannian manifold $(\mathcal{M}, g)$ with variable curvature $\kappa(x) \leq -\kappa_0 < 0$, as well as a discrete topological connectivity graph $\mathcal{T}$. Viewing the attention mechanism as an Ising-like system, each pair (i, j) in $\mathcal{T}$ contributes an interaction $S_i S_j$ modulated by the manifold geometry. Let $\Delta_{\mathcal{M}}$ be the Laplace–Beltrami operator. Under fractal dimension δ , the *near-zero* density behaves like $\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1}$. However, the effective gap ξ_{crit} and the product $\beta \lambda_{\text{max}}(\mathcal{T})$ (where $\lambda_{\text{max}}(\mathcal{T})$ is the spectral radius of the adjacency) collectively set the threshold at which $\text{Var}_{\beta}(H)$ diverges.

Proposition 1 (General Approximate Formula for β_c under Curvature & Topology). *Let $\kappa(x) \leq -\kappa_0 < 0$ on $\mathcal{M}$, and let $\lambda_{\text{max}}(\mathcal{T})$ denote the maximal eigenvalue of the connectivity adjacency matrix. Assume near $\xi = 0$ we have $\rho(\xi) \approx C |\xi|^{\frac{\delta}{2}-1}$. Then, for an effective radius r_{eff} ,*

$$\beta_c(\delta, \kappa, \mathcal{T}) \approx [\lambda_{\text{max}}(\mathcal{T})]^{-1} \exp\left(C(\kappa_0) \delta r_{\text{eff}}\right),$$

where $C(\kappa_0)$ is a constant determined by how the negative curvature influences small-eigenvalue localization. In the special case $\kappa_0 = 1$ and $\lambda_{\text{max}}(\mathcal{T}) = 1$ (fully connected or mean-field), we recover $\beta_c(\delta) \sim e^{2\delta r_{\text{eff}}}$.

Sketch (Mostly in Equations):

$$(1) \text{ Define } H = - \sum_{\langle i, j \rangle \in \mathcal{T}} S_i S_j \text{ on manifold with } \kappa(x) \leq -\kappa_0.$$

$$(2) \text{ Partition function: } Z(\beta) = \sum_{\{S\}} \exp\left(\beta \sum_{\langle i, j \rangle \in \mathcal{T}} S_i S_j\right).$$

$$(3) \text{ Critical condition: } \frac{\partial^2}{\partial \beta^2} \log Z(\beta) \Big|_{\beta=\beta_c} \rightarrow \infty \Leftrightarrow \text{Var}_{\beta}(H) \rightarrow \infty.$$

- (4) Near $\xi = 0$: $\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1}$.
- (5) Curvature modifies $\xi_{\text{crit}} \sim \exp(-C(\kappa_0) \delta r_{\text{eff}})$.
- (6) Topology factor: $\beta \lambda_{\text{max}}(\mathcal{T}) \xi_{\text{crit}} \approx 1$.
- $\therefore \beta_c(\delta, \kappa, \mathcal{T}) \approx [\lambda_{\text{max}}(\mathcal{T})]^{-1} \exp(C(\kappa_0) \delta r_{\text{eff}})$.

9 Numerical Experiments

9.1 Experimental Setup

We evaluate our hyperbolic (and, more generally, curved) attention approach on synthetic hierarchical data, as well as standard tasks in language modeling. Let $\{x_i\}_{i=1}^N$ be input tokens. We compute both the standard dot-product attention (eq. (2)) and the extended curved attention. We track E_{diss} and measure test perplexity or classification accuracy. We vary topological adjacency $\mathcal{T}$ (from fully-connected to sparser structures) and curvature scale $\kappa_0 \in \{0.5, 1.0, \dots\}$, comparing the empirical β_c to eq. (1)’s predictions.

9.2 Results and Interpretations

Energy Minimization. We observe that E_{diss} is consistently minimized close to the predicted $\beta_c(\delta, \kappa, \mathcal{T})$, confirming that topological constraints and curvature lead to consistent shifts in critical thresholds.

Performance Gains. When $\delta > 0$ and curvature is sufficiently negative ($\kappa(x) < 0$), tasks with large hierarchical or tree-like data exhibit improved perplexity and higher classification accuracy. Gains are pronounced if $\mathcal{T}$ is chosen to mirror data adjacency structures.

10 Criticality and Correlation Length under Hyperbolic Embedding Setup

In our construction, the attention mechanism embeds the query and key vectors into a hyperbolic manifold via a smooth map:

$$S_i := \phi((XW_Q)_i), \quad S_j := \phi((XW_K)_j),$$

and defines a hyperbolic attention score via:

$$A_{ij}^{(\mathbb{H})} = f_{\kappa}(\langle S_i, S_j \rangle_{\mathbb{H}}),$$

where $\langle \cdot, \cdot \rangle_{\mathbb{H}}$ is the hyperbolic inner product.

We interpret $\langle S_i, S_j \rangle_{\mathbb{H}}$ analogously to the spin coupling term $S_i S_j$ in classical statistical mechanics. Therefore, the system’s effective Hamiltonian becomes:

$$H = - \sum_{\langle i, j \rangle} \langle S_i, S_j \rangle_{\mathbb{H}}.$$

This resembles a vector-spin model (such as the $O(N)$ or Heisenberg model), with spins living on a curved manifold.

The key insight is that, near the critical point, the correlation function of the spins (hyperbolic embeddings) behaves like the Green’s function of the Laplace–Beltrami operator on the manifold:

$$\langle S_i \cdot S_j \rangle_\beta \approx G(x_i, x_j; \beta) \sim ((-\Delta_{\mathbb{H}} + m_\beta^2)^{-1})(x_i, x_j),$$

where m_β^2 is a mass parameter that tends to zero at criticality. The correlation length $\xi(\beta)$ diverges as:

$$\xi(\beta) \sim \frac{1}{m_\beta}, \quad \text{with} \quad m_\beta^2 \sim |\beta_c - \beta|.$$

At criticality, we have $m_\beta \rightarrow 0$, implying:

$$\xi_{\text{crit}} \rightarrow \infty.$$

Since β is the control parameter responsible for this divergence, it sets the inverse of the emergent scale in the system. Thus, the critical inverse temperature must satisfy:

$$\boxed{\beta_c \sim \frac{1}{\xi_{\text{crit}}}}.$$

This relationship remains valid even in the vector embedding setting, provided that pairwise hyperbolic inner products $\langle S_i, S_j \rangle_{\mathbb{H}}$ play the role of alignment interactions across the attention manifold.

11 Alternative Coupling: Setting $J_{ij} \propto \langle S_i, S_j \rangle_{\mathbb{H}}$

We now consider an alternative physical interpretation, where the hyperbolic inner product determines the **coupling strength** J_{ij} between spins or nodes, rather than being interpreted as the spin interaction term itself.

Geometric Coupling Strengths

Specifically, we write the Ising-like Hamiltonian in the form:

$$H = - \sum_{\langle i, j \rangle} J_{ij} \cdot \sigma_i \sigma_j,$$

and now define:

$$J_{ij} := \gamma \cdot \langle S_i, S_j \rangle_{\mathbb{H}},$$

where:

- $\sigma_i, \sigma_j \in \{-1, +1\}$ are scalar Ising spins,
- $S_i := \phi((XW_Q)_i)$, $S_j := \phi((XW_K)_j) \in \mathbb{H}^n$ are hyperbolically embedded query/key projections,
- $\gamma \in \mathbb{R}$ is a global scaling parameter (e.g., learned or fixed).

This configuration distinguishes between the spin variables σ_i , which are scalar degrees of freedom or simplified hidden states, and the hyperbolic embeddings S_i , which modulate how strongly these degrees of freedom interact, via geometry.

Interpretation in Transformer Architectures

In the context of transformers, potential candidates for the spin-like degrees of freedom σ_i include:

1. **Attention gate activations** (binary approximations during training),
2. **Head masking variables** in sparse or pruned attention,
3. **Latent binary decisions** from discrete bottlenecks (e.g., in VQ-VAE or L0 gates),
4. **Binary pattern selectors** in mixture-of-experts or routing mechanisms.

Meanwhile, the embeddings $S_i = \phi((XW_Q)_i)$ and $S_j = \phi((XW_K)_j)$ still denote the query and key projections embedded in a curved (e.g., hyperbolic) latent space.

Implication for Critical Behavior

The hyperbolic inner product $\langle S_i, S_j \rangle_{\mathbb{H}}$ now determines the geometry-aware coupling J_{ij} , meaning that spins interact more strongly when their attention embeddings are geometrically aligned in hyperbolic space. The divergence of the correlation length at criticality now results from:

$$\beta_c \cdot J_{ij} \cdot \xi_{\text{crit}} \sim 1.$$

Solving this gives:

$$\beta_c \sim \frac{1}{\langle S_i, S_j \rangle_{\mathbb{H}} \cdot \xi_{\text{crit}}}.$$

Assuming normalized embeddings or uniform scaling of $\langle S_i, S_j \rangle_{\mathbb{H}} \sim \text{const}$, we recover:

$$\boxed{\beta_c \sim \frac{1}{\xi_{\text{crit}}}}.$$

Thus, the scaling behavior of critical inverse temperature remains consistent with the previous setup, but the interpretation of what drives the coupling—geometry vs. spin—shifts the modeling perspective.

12 Unified Model: Hyperbolic Spins and Geometry-Modulated Couplings

We now consider a comprehensive geometric-statistical framework in which:

1. The **spin variables** $S_i \in \mathbb{H}^n$ are themselves hyperbolic vectors representing the transformer queries and keys after nonlinear embedding.
2. The **interaction energy** is constructed from a hyperbolic inner product, which both acts as a spin-spin alignment term and modulates the coupling strength.

Hamiltonian Structure

We define the Hamiltonian:

$$H = - \sum_{\langle i,j \rangle} J_{ij} \cdot \langle S_i, S_j \rangle_{\mathbb{H}},$$

where:

$$J_{ij} := \gamma \cdot \langle S_i, S_j \rangle_{\mathbb{H}}. \quad (20)$$

Hence, the interaction term becomes:

$$H = -\gamma \sum_{\langle i,j \rangle} \langle S_i, S_j \rangle_{\mathbb{H}}^2.$$

This model leads to a *quadratic dependence* on the hyperbolic inner product, reminiscent of continuous-spin models (e.g., the XY model generalized to curved spaces).

Interpretation in Transformer Context

Here, both the values $S_i = \phi((XW_Q)_i)$ and $S_j = \phi((XW_K)_j)$ are attention embeddings embedded in hyperbolic space. This reflects full geometric awareness of the latent structure. Examples of such hyperbolic variables in a transformer might include:

- **Query and Key vectors** under non-Euclidean embeddings (e.g., wrapped by tanh, exponential map, or Möbius projection),
- **Token representations** in models trained with hyperbolic loss functions,
- **Contextual position encodings** when modeled as points on $\mathbb{H}^n$.

Implications for Critical Scaling

The effective correlation structure depends on the second moment of the hyperbolic inner product:

$$\langle \langle S_i, S_j \rangle_{\mathbb{H}}^2 \rangle.$$

The divergence of the correlation length at criticality is governed by when:

$$\beta_c \cdot \gamma \cdot \langle S_i, S_j \rangle_{\mathbb{H}}^2 \cdot \xi_{\text{crit}} \sim 1.$$

Solving for the critical inverse temperature gives:

$$\beta_c \sim \frac{1}{\gamma \cdot \langle S_i, S_j \rangle_{\mathbb{H}}^2 \cdot \xi_{\text{crit}}}.$$

Assuming embeddings are approximately normalized or stabilized such that $\langle S_i, S_j \rangle_{\mathbb{H}}^2 \sim \text{const}$, we again recover:

$$\boxed{\beta_c \sim \frac{1}{\xi_{\text{crit}}}},$$

preserving consistency with the spectral theory viewpoint, while now attributing the statistical interaction and geometric modulation to the same underlying object.

Remark on Learning Dynamics

In practical transformer implementations, this joint structure suggests that attention updates—and even training convergence—may reflect emergent criticality tied to the geometry of latent embeddings. The system may self-tune toward a scale-free regime where:

$$\text{Var}_\beta(H) \rightarrow \infty,$$

coinciding with high expressivity and long-range dependencies in learned representations.

13 From Correlation Length to Geometric-Fractal Critical Scaling

We now explain how the foundational identity

$$\boxed{\beta_c \sim \frac{1}{\xi_{\text{crit}}}} \tag{21}$$

logically leads to the full expression

$$\beta_c(\delta, \kappa, \mathcal{T}) \approx [\lambda_{\max}(\mathcal{T})]^{-1} \exp(C(\kappa_0) \delta r_{\text{eff}}). \tag{22}$$

Step 1: Interpreting ξ_{crit}

The correlation length ξ_{crit} refers to the spatial scale over which statistical dependencies between spin variables (or attention embeddings) persist. It is well known in statistical physics and quantum field theory that the two-point function decays exponentially as:

$$\langle S(x)S(y) \rangle \sim e^{-d(x,y)/\xi_{\text{crit}}},$$

where $d(x, y)$ is the geodesic distance.

At the spectral level, this exponential decay is governed by the **smallest nonzero eigenvalue** $\xi_{\min}$ of a Laplace-type operator (e.g., the Laplace–Beltrami operator), with:

$$\xi_{\text{crit}} \sim \frac{1}{\sqrt{\xi_{\min}}}. \tag{23}$$

Hence, in the critical regime where the mass-like term vanishes, the divergence of correlation length is equivalent to:

$$\boxed{\beta_c \sim \frac{1}{\xi_{\text{crit}}}} \iff \beta_c \sim \sqrt{\xi_{\min}}.$$

Step 2: How δ Enters via Spectral Geometry

Now consider that the system is embedded in a **negatively curved manifold** (e.g., $\mathbb{H}^n$) with boundary at infinity. The Laplace–Beltrami operator $\Delta_{\mathcal{M}}$ has spectral density near $\xi \approx 0$ that depends on the geometry and topology of the space.

In particular, when the boundary at infinity carries a **fractal limit set** of dimension δ , spectral geometry (e.g., Patterson–Sullivan theory, Lax–Phillips scattering theory) implies:

$$\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1} \quad \text{as } \xi \rightarrow 0. \tag{24}$$

This fractal dimension δ thus controls the **accumulation rate of small eigenvalues**, i.e., how “dense” the spectrum is near zero.

Step 3: Estimating ξ_{crit} from δ and κ

From heat kernel estimates and geometric scattering theory in hyperbolic spaces, one obtains:

$$\xi_{\text{crit}} \sim \exp(-C(\kappa_0) \delta r_{\text{eff}}), \quad (25)$$

where:

- δ is the dimension of the fractal limit set at infinity,
- r_{eff} is the effective radius of the geometry,
- $C(\kappa_0)$ is a curvature-dependent constant (larger for stronger curvature).

This is justified by noting that the eigenvalues of $\Delta_{\mathbb{H}^n}$ on spaces with fractal boundary structure become **exponentially small** as the complexity of the boundary increases.

Step 4: Incorporating Topological Coupling

In practice, the attention mechanism or statistical system has coupling strengths mediated by an adjacency graph $\mathcal{T}$. The influence of topology appears via the **spectral radius** $\lambda_{\text{max}}(\mathcal{T})$, which rescales the energetic contributions.

The correct divergence condition for the variance of the Hamiltonian becomes:

$$\beta_c \cdot \lambda_{\text{max}}(\mathcal{T}) \cdot \xi_{\text{crit}} \sim 1,$$

implying:

$$\boxed{\beta_c(\delta, \kappa, \mathcal{T}) \approx [\lambda_{\text{max}}(\mathcal{T})]^{-1} \exp(C(\kappa_0) \delta r_{\text{eff}}).}$$

Conclusion

Therefore, even though:

- β_c is defined via a statistical mechanics partition function over the bulk manifold;
- δ is defined via the boundary limit set (fractal boundary geometry);

they are deeply linked through **spectral geometry**, because the eigenvalue accumulation near $\xi = 0$ reflects both the curvature and the fractal nature of the boundary. The final formula (22) is thus a geometrically and physically justified refinement of the spectral criticality condition $\beta_c \sim 1/\xi_{\text{crit}}$.

14 Connections to Langlands Program and Quantum Security

14.1 Langlands Correspondence and Encryption

Recent frameworks incorporate automorphic forms and Galois representations for post-quantum cryptography. By embedding W_Q, W_K in automorphic forms derived from $\mathbb{H}^n/\Gamma$, one can exploit modular properties to secure training and inference.

Theorem 1 (Langlands-based Security). *Suppose the hyperbolic manifold $\mathbb{H}^n$ admits a discrete group Γ with corresponding automorphic forms $\{\Phi_k\}$. Mapping attention weights into $(W_Q, W_K) \mapsto (\Phi_{k_1}, \Phi_{k_2})$ yields a post-quantum encryption layer, provided that the underlying Galois representations remain unknown to adversaries.*

14.2 Quantum Security and Hyperbolic Geometry

Lorentz transformations and continuous curvature re-parameterizations can re-map latent spaces away from known adversarial coordinates. Coupled with Langlands-based encryption, this approach complicates quantum-based attacks on parameter states.

15 Further Extensions

15.1 Lorentz-based Real-time Adaptation

We may augment eq. (3) with a Lorentz transform Λ acting on $\phi((XW_Q)_i)$ in real-time:

$$\phi((XW_Q)_i) \mapsto \Lambda \phi((XW_Q)_i), \quad \Lambda^T g_M \Lambda = g_M, \quad (26)$$

where g_M is the Minkowski metric. This allows the system to adapt to distribution shifts without retraining from scratch.

16 Conclusion and Open Problems

We have presented a 20-page, rigorously mathematical treatment of hyperbolic attention mechanisms at critical scaling, *then extended it to a more general framework where curvature, topology, and fractal dimension all influence $\beta_c(\delta, \kappa, \mathcal{T})$* . Our derivations show that $\beta_c(\delta)$ (or $\beta_c(\delta, \kappa, \mathcal{T})$) admits a closed-form or approximate dependence on these parameters, ensuring minimal energy dissipation at criticality. Future research directions include extending the spectral feedback approach to larger networks, exploring real-time Lorentz transformations, investigating topological reconfiguration, and deepening connections to advanced number-theoretic frameworks (Langlands, etc.).

References

- [1] W. H. Chuang, *Energy Dynamics in Self-Attention: Theoretical Insights and Alternatives to Standard Scaling*. Preprint, 2024.
- [2] W. H. Chuang, *Hausdorff Dimension of Well-Distributed Schottky Groups and Generalizations to Higher-Dimensional Hyperbolic Spaces*. Preprint, 2025.
- [3] A. Vaswani et al., *Attention is All You Need*. In NIPS, 2017.

A Proofs of Technical Lemmas

Lemma 1 (Spectral Identity 3.2). *Let $\Delta_{\mathbb{H}}$ be the Laplace–Beltrami operator on $\mathbb{H}^n$. Suppose its spectral density near $\xi = 0$ behaves as $\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1}$ under fractal dimension δ . Then for $\xi \rightarrow 0$:*

$$\int_0^\epsilon |\xi|^{\frac{\delta}{2}-1} d\xi < \infty \quad \Longleftrightarrow \quad \delta > 0.$$

Extended Proof. Let $p = \frac{\delta}{2} - 1$ so that the integrand near $\xi = 0$ is $|\xi|^p$. To check for convergence at the lower limit $\xi = 0$, we examine

$$\int_0^\epsilon |\xi|^p d\xi.$$

A standard one-dimensional improper integral test tells us that

$$\int_0^\epsilon x^p dx = \begin{cases} \frac{\epsilon^{p+1}}{p+1}, & \text{if } p \neq -1, \\ \log(\epsilon), & \text{if } p = -1. \end{cases}$$

We need this integral to be finite as $\epsilon \rightarrow 0$. It is well known that

$$\int_0^\epsilon x^p dx < \infty \iff p > -1.$$

Here, $p = \frac{\delta}{2} - 1$. Rewriting $p > -1$ gives

$$\frac{\delta}{2} - 1 > -1 \iff \frac{\delta}{2} > 0 \iff \delta > 0.$$

Hence, the integral converges if and only if $\delta > 0$. As a consequence, the spectral density $\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1}$ is integrable near $\xi = 0$ precisely when the fractal dimension δ is positive. This completes the proof. $\square$

B Supplementary Mathematical Derivations

Proposition 2 (Critical Divergence — Equation (15)). *Under the spectral-density assumption (5), the second derivative test of the free energy (or log-partition function) (8) diverges exactly at*

$$\beta_c(\delta) \sim e^{2\delta r_{\text{eff}}}.$$

Further imposing $\delta = \frac{\log(2m-1)}{r_{\text{eff}}}$ yields

$$\beta_c(\delta) \sim (2m-1)^2.$$

Extended Proof. We sketch the main ideas in several steps:

Step 1: Setup of the Partition Function and Hamiltonian. Consider the (hyperbolic) attention mechanism in a statistical-mechanical analogy, where the total energy (Hamiltonian) takes an Ising-like form:

$$H = - \sum_{\langle i,j \rangle} S_i S_j,$$

and the partition function is given by

$$Z(\beta) = \sum_{\{S_i\}} e^{-\beta H}.$$

Here, β plays the role of an inverse temperature. Criticality is signaled by the divergence of the second derivative of $\log Z(\beta)$ w.r.t. β , i.e.,

$$\frac{\partial^2}{\partial \beta^2} \log Z(\beta) = \text{Var}_\beta(H) = \langle H^2 \rangle_\beta - \langle H \rangle_\beta^2.$$

Hence, establishing that $\text{Var}_\beta(H)$ blows up at $\beta = \beta_c(\delta)$ identifies the critical point.

Step 2: Relation to Spectral Density. In hyperbolic geometry $\mathbb{H}^n$, one connects pairwise spin correlations to the Laplace–Beltrami operator $\Delta_{\mathbb{H}}$ and its eigenvalue spectrum. Let $\{\xi\}$ denote the eigenvalues of $-\Delta_{\mathbb{H}}$. If near $\xi = 0$ the density of states behaves like

$$\rho(\xi) \sim |\xi|^{\frac{\delta}{2}-1},$$

then low-lying (near-zero) modes become critical when $\beta \xi$ reaches a threshold.

Step 3: Divergence Condition. We typically see that near $\beta = \beta_c$, the correlation length diverges and the variance $\text{Var}_{\beta}(H)$ has a contribution

$$\int_0^{\xi_{\max}} \frac{\rho(\xi)}{(1 - \beta \xi)^2} d\xi,$$

which diverges if $\beta \xi_{\text{crit}} = 1$ with $\xi_{\text{crit}} \approx 0$. If $\rho(\xi) \sim |\xi|^p$ near zero, the integral diverges exactly at that threshold.

Step 4: Identifying $\beta_c(\delta)$. We deduce that $\frac{1}{\beta_c(\delta)} \approx \xi_{\text{crit}} \sim e^{-2\delta r_{\text{eff}}}$, hence

$$\beta_c(\delta) \sim e^{2\delta r_{\text{eff}}}.$$

Step 5: Substituting $\delta = \frac{\log(2m-1)}{r_{\text{eff}}}$. Finally, $\delta = \frac{\log(2m-1)}{r_{\text{eff}}}$ yields

$$\beta_c(\delta) \sim (2m-1)^2,$$

hence eq. (16) follows. This completes the proof. $\square$

Note: By further adopting a curvature parameter κ_0 and topological adjacency $\mathcal{T}$, one generalizes to the expression

$$\beta_c(\delta, \kappa, \mathcal{T}) \approx [\lambda_{\max}(\mathcal{T})]^{-1} \exp(C(\kappa_0) \delta r_{\text{eff}}),$$

as discussed in Proposition 1.